

Product Concepts & Intended Uses – Parkinson’s Disease Treatment

Portfolio Selection: Portfolio 3 – PK1 protein biomarker for Parkinson’s Disease (PD)

Product Concepts:

- **Therapeutic:** Biologic drug to downregulate and/or inhibit PK1 expression.
- **Device:** Immunoassay analyzer for detection of PK1 concentration in human blood plasma.

1) Product Classification:

- **Therapeutic:** Biologic
- **Device:** Class II medical device regulated by the CBER

2) Target Disease Area:

Parkinson’s Disease. Therapeutic treatment would be ideal for later stages of PD but might be suitable for those with early signs (i.e. increasing PK1 levels) to slow the onset of PD before symptoms occur.

3) Standard of Care:

- **Therapeutic:** Current care for patients with Parkinson’s disease mostly aims to reduce severe symptoms associated with Parkinson’s. A common prescription drug for this is carbidopa and levodopa to treat common symptoms of late PD onset, including stiffness, difficulty moving, and tremors.
- **Device:** No current definitive diagnostic test exists. PD diagnosis is primarily based on symptoms, exclusion of other causes for symptoms, and brain scans.

4) Unmet Need:

- **Therapeutic:** Solution to treat patients with EARLY onset of PD (i.e., someone who is at risk of contraction or has a familial history of PD).
- **Device:** No current diagnostic or screening device exists. Product meets need for diagnostic and screening tools.

5) How will your product be used?

Levels of Specificity for **therapeutic:** AAV to deliver genetic code for an siRNA to reduce the translation of PK1.

- **Identification of...**
 - **function:** Incorporate a gene into the genome on C5 of an siRNA to stop the expression of PK1 mRNA
 - **tissue type:** Neurons
 - **disease entity or target population:** Early-Stage PD; When PK1 levels are >450 +/- 100 units and level out prior to symptoms. “The PK1 biomarker can be measured and differentiate patients with onsets of the disease up to

2 years prior to physical manifestation of symptoms, as levels of PKS1 in the blood plasma increase over the period (24 months)”.

- effect on clinical outcome: Use of this therapeutic decreases the levels of PKS1 protein in neurons.

Levels of Specificity for diagnostic medical device:

- **Identification of...**

- Identification or measurement of a physical parameter (e.g., image, heart rate) or biochemical parameter (e.g., analyte): PKS1
- a new or specific target population (e.g., women, children of a certain age range) or anatomical location (e.g., MR of the brain)
 - Adults 50 years and older with PD symptoms.
 - Rationale: Diagnostic tool, PD typically affects those 50+
 - Adults 50 years and older without PD symptoms.
 - Rationale: Screening tool for early detection and treatment
 - Adults under 50 years without symptoms who have elevated risk of developing PD.
 - Rationale: Screening tool for early detection and treatment
 - Adults under 50 years with symptoms
 - Rationale: Diagnostic tool for early-onset PD
- the clinical use of the measurement: Diagnosis and screening for PD
- an effect on the clinical outcome: Screening for PD reduces time to treatment thereby slowing disease progression. Diagnosing PD guides therapeutic decisions resulting in decreasing symptoms.

Intended Use Statements

Therapeutic:

The therapeutic will be a biologic drug designed to downregulate and/or inhibit the expression of PKS1 via gene therapy. It is intended to treat those with early signs of Parkinson’s disease by aiming to slow the disease’s onset before symptoms occur and become prominent. The effectiveness of this biologic will be evaluated by the medical device product to determine patient treatment plans going forward.

Device:

The device is an immunoassay analyzer designed for the quantitative determination of PKS1 protein levels in human blood plasma. It is intended for the diagnosis and screening of Parkinson’s Disease in a clinical laboratory setting for professional use only. Diagnostic application of this device is intended to be used in conjunction with the therapeutic product to establish and implement an appropriate treatment plan.